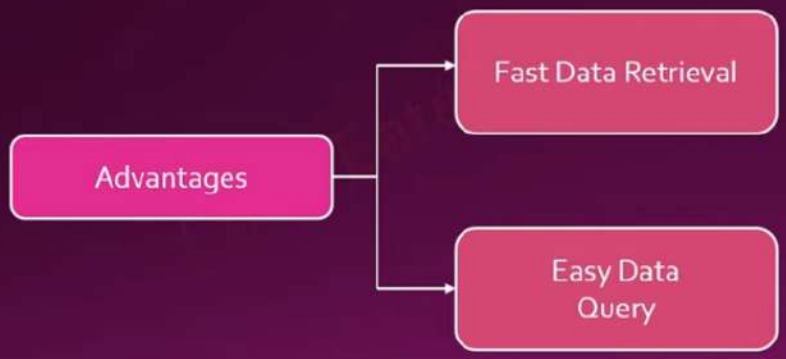
By Code Eater

The Graph

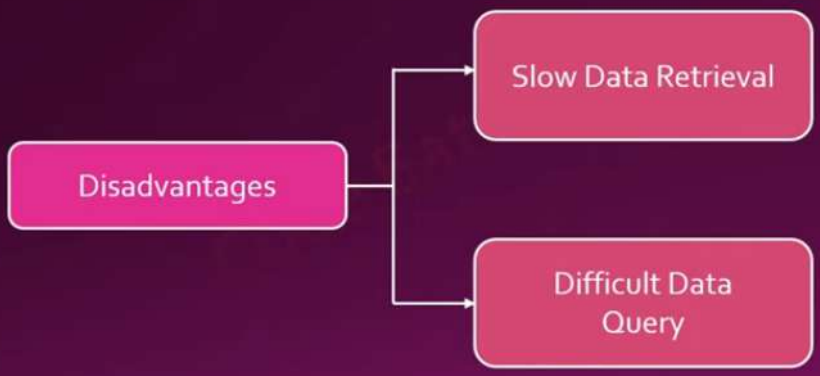
Web2



Web2

Web3

Data


Disadvantages
Slow Data Retrieval
Difficult Data Query

Two Solution

Centralized ⇨ DataStax

Decentralized ⇨ The Graph

DataStax
Data
DATASTAX
Data

The Graph

- **The Graph** is an indexing protocol and decentralized network that enables developers to efficiently access and retrieve data from blockchains. It aims to provide an infrastructure layer for Web3 applications by organizing and making blockchain data easily accessible and searchable.
- An **indexing protocol** refers to a set of rules, standards, and methodologies that define how data is organized, stored, and made searchable for efficient retrieval. It provides a framework or protocol for creating and maintaining indexes, which are data structures designed to facilitate faster access and retrieval of specific information.

The Graph Protocol — super easy Hinglish explain 🔥

Socho **blockchain = ek bahut bada register** 📒
Har transaction, event, data **line by line likha hota hai**.

👉 **Problem:**
Blockchain se data nikalna **direct bahut slow & complex** hota hai.
Imagine: *"Last 10 swaps batao"* — tumhe poora register padhna padega 😵

👉 **Solution:**

The Graph

Ye blockchain ka **Google** hai 🔍

🧠 Simple real-life example

Without The Graph ❌

- Tum library jaate ho
- Har book ka **har page** khud read karte ho
- Answer milta hai… 2 ghante baad 🥱

With The Graph ✅

- Librarian ne **index bana rakha hai**
- Tum bolo: *"Uniswap swaps last 24 hours"*
- Turant jawab ⚡

➡️ **The Graph = Librarian + Index system**

🧩 Core Concepts (feel ke saath)

1️⃣ Subgraph kya hota hai?

Subgraph = **rule book** 📘

Tum bolte ho:

> "Mujhe ye-ye events chahiye, ye-ye data store karo"

Example:

- Swap event
- Transfer event
- User address
- Amount

📌 Ye sab ek **Subgraph** me define hota hai

2️⃣ Indexing ka matlab

Blockchain pe jo bhi event hota hai:

```vbnet
Event → The Graph dekhta hai → Data structured format me store
```

Jaise:

```powershell
Raw blockchain data ❌
↓
Clean table-like data ✅
```

3️⃣ Query kaise hoti hai?

Tum frontend se bolte ho:

graphql Copy code

```graphql
{
  swaps(first: 10) {
    user
    amount
    timestamp
  }
}
```

👉 Tumhe **direct answer milta hai**, blockchain read nahi karna padta 🤯

🔥 Why Web3 projects NEED The Graph

Without it:

- Slow dApps 🐢
- Complex backend
- Direct blockchain calls = pain

With it:

- Fast frontend ⚡
- Simple queries
- Clean data
- Production-ready apps 🚀

🧪 Real example (feel it)

Uniswap UI

- "Top tokens"
- "Recent swaps"
- "Volume 24h"

❓ Kya Uniswap har baar blockchain read karta hai?
❌ NO

✅ **They use The Graph**

🧠 One-line summary (yaad rakhne ke liye)

> **The Graph blockchain ka Google hai — jo slow raw data ko fast searchable data bana deta hai.**

The Graph Components

1. Subgraph Definition Language (SDL)

Developers define subgraphs using SDL, a schema language specifically designed for blockchain data. A subgraph represents a subset of data from a specific blockchain network that developers want to index and make available for querying.

2. Graph Node

The Graph Node is software that processes and indexes data from blockchains based on the defined subgraphs. It acts as a bridge between the blockchain and the query layer.

3. Query Layer

Developers interact with The Graph's query layer to retrieve data from the indexed subgraphs. They can use GraphQL, a flexible query language, to express the data they need and receive responses in a structured format.

🎥 Big Picture (ek line me)

The Graph =

👉 *Blockchain ka raw data → clean, fast, searchable data*

1️⃣ Subgraph Definition Language (SDL)

🧠 ***"Tum kya dekhna chahte ho?"***

Real-life feel 👇

Soch tu **CCTV control room** ka owner hai.

Camera har second **sab kuch record** kar raha hai:

- log aa rahe
- ja rahe
- lad rahe
- chai pee rahe ☕

Par tujhe sirf chahiye:

> "Sirf **gate entry logs + time + person ID**"

👉 **SDL = tumhara instruction paper**

Blockchain context me

SDL me tu bolta hai:

- Kaunsa **contract**
- Kaunsa **event**
- Kaunsa **data store** karna hai

Example (feel):

graphql Copy code

```graphql
type Swap {
  id: ID!
  user: Bytes!
  amountIn: BigInt!
  amountOut: BigInt!
  timestamp: BigInt!
}
```

📌 Matlab:

> "Mujhe sirf swap ka data chahiye, baaki blockchain ka kachra nahi"

🔥 **SDL = filter + structure**

2️⃣ Graph Node

⚙️ *"Kaam karne wala mazdoor"*

Real-life feel 👇

Ab tu CCTV instructions de chuka hai.

Ab kaun kaam karega?

👉 **Security guard + computer system**

Woh:

- CCTV dekhta rahega 👀
- Tera rule follow karega
- Important cheez likh lega register me 📒

Blockchain context me

Graph Node kya karta hai?

- Blockchain continuously read karta hai
- SDL ke rules follow karta hai
- Event aaya? → Data extract
- Structured DB me store

Flow:

```powershell
Blockchain
   ↓
Graph Node (filter + process)
   ↓
Indexed Data (clean)
```

📌 Isliye likha hota hai:

> *"Graph Node acts as a bridge"*

🔥 **Graph Node = brain + worker**

3️⃣ Query Layer

🔍 *"Ab sawal pucho"*

Real-life feel 👇

Ab tu control room me aata hai aur bolta hai:

> "Kal raat 10 baje ke baad kaun kaun aaya?"

Security guard:
❌ "Puri CCTV tape dekh lo"
✅ "Ye lo list sir 👇"

Blockchain context me

Tu frontend se **GraphQL** query likhta hai:

graphql Copy code

```graphql
{
  swaps(first: 5, orderBy: timestamp, orderDirection: desc) {
    user
    amountIn
    amountOut
  }
}
```

⚡ Result:

- Fast
- Clean
- Exact wahi jo chahiye

🔥 **Query Layer = search box**

🧩 Ab teeno ko ek saath feel karo

🎯 One full flow story

1. **SDL** → *"Mujhe kya data chahiye"*
2. **Graph Node** → *"Main jaa ke dhundh ke rakhunga"*
3. **Query Layer** → *"Ab puch lo jo chahiye"*

Diagram feel:

```scss
Tum (Developer)
   ↓ (SDL)
Graph Node
   ↓
Indexed Data
   ↓ (GraphQL)
Tumhara Frontend
```

💥 One killer analogy (yaad rahegi)

> **Blockchain = Raw CCTV footage**
> **SDL = Instructions**
> **Graph Node = Guard + System**
> **Query Layer = Search bar**

🧠 Exam / Interview one-liner

> *"The Graph lets developers define what blockchain data to index (SDL), processes it via Graph Nodes, and exposes it through a GraphQL query layer for fast access."*

Subgraph

from
to amount

```
event TransferEvent(address from,address to,uint amount);

function transfer(address payable to)  public payable{
    to.transfer(msg.value);
    emit TransferEvent(msg.sender,to,msg.value);
}
```

from	to	amount
0xabc	0x123	10
0xecf	0xdef	500
0xdab	0x4ab	30

Subgraph

- In the context of the graph protocol in blockchain, a subgraph refers to a subset of data from the blockchain that is organized and indexed for efficient querying and analysis.
- Indexing refers to the process of creating data structures that organize the data in a manner that enables faster searching and retrieval. In the context of the graph protocol, indexing involves creating indexes or data structures that optimize the querying of specific data within the subgraph.
- By using indexes, queries can quickly locate the desired data within the subgraph without the need to scan the entire dataset. This significantly improves the performance of queries and reduces the time it takes to retrieve information.
- Indexes can take various forms, such as hash tables, B-trees, or other data structures optimized for quick data access. The specific indexing techniques employed depend on the requirements of the subgraph and the nature of the data being indexed.

🧠 Big One-Line First (mind lock 🔒)

> Subgraph = Blockchain ke jungle se sirf useful data nikaal ke, table form me rakh dena — taaki future me instantly mil jaaye.

Aur ye sab hota hai The Graph ke through.

🟣 IMAGE 1 — Code → Events → Table (MAGIC ✨)

Solidity code (left side)

```solidity
event TransferEvent(address from, address to, uint amount);

function transfer(address payable to) public payable {
  to.transfer(msg.value);
  emit TransferEvent(msg.sender, to, msg.value);
}
```

FEEL IT 👇

Soch:

- Har baar `transfer()` call hota hai
- Blockchain chillata hai: "TRANSFER HUA!" 📢
- Saath me bolta hai:
  - from = kaun
  - to = kaun
  - amount = kitna

⛓️ Blockchain reality

Blockchain ke liye ye sab:

- logs hain
- unordered
- raw
- unreadable for frontend

❌ Direct use karna = pain

🔥 Yahin Subgraph aata hai

Subgraph bolta hai:

> "Jab bhi `TransferEvent` aaye
> 👉 mujhe `from`, `to`, `amount` nikaal ke
> 👉 ek clean table me daal dena"

Result (right side table)

from	to	amount
0xabc	0x123	10
0xecf	0xdef	500
0xdab	0x4ab	30

💥 **THIS is indexing**

🧠 IMAGE 2 — Indexing ka actual matlab (feel ke saath)

🔍 Line 1 explained

> *Subgraph refers to a subset of data from the blockchain*

FEEL:

Blockchain = **poora universe** 🌍
Subgraph = **sirf ek city ka Google Map** 🗺️

Tum bolte ho:

> "Mujhe sirf TRANSFER ka data chahiye, poora universe nahi"

⚙️ Line 2 — Indexing kya hota hai?

> *Indexing = organizing data for faster searching*

Real-life analogy 🔥

Phone contacts 📱:

- Naam ke bina number dhundhna = nightmare
- Naam ke saath = instant

👉 Naam = **index**

Blockchain me:

Raw blockchain:

```mathematica
Block → Tx → Log → Event → Data
```

Indexed subgraph:

```pgsql
TransferEntity {
  from
  to
  amount
  timestamp
}
```

✳️ Difference = search speed

🚀 Line 3 — "scan entire dataset" kyun nahi?

Without indexing:

- Poora blockchain scan ❌
- 10,000 blocks ❌
- Slow UI ❌

With subgraph:

- Direct table lookup ✅
- Milliseconds response ⚡

🌳 Line 4 — Hash tables, B-trees (simple feel)

Tu exam me ye bolega:

> "Indexes are internal data structures used for fast lookup."

But FEEL ye hai 👇

Hash table:

- Address → direct data
- Like JS object `{}`

B-tree:

- Sorted data
- Range queries (last 24 hours, top 10, etc.)

👉 Ye sab Graph Node internally sambhalta hai

Tumhe tension lene ki zarurat nahi 😊

🧩 Ab poora flow ek baar FEEL karo

```scss
User calls transfer()
        ↓
Blockchain emits event
        ↓
Graph Node listens 👂
        ↓
Subgraph rules apply (SDL)
        ↓
Data indexed (table)
        ↓
Frontend queries via GraphQL
        ↓
Fast result ⚡
```

🔥 ONE DEADLY REAL EXAMPLE (Uniswap feel)

Uniswap UI shows:

- Recent swaps
- Top traders
- Volume

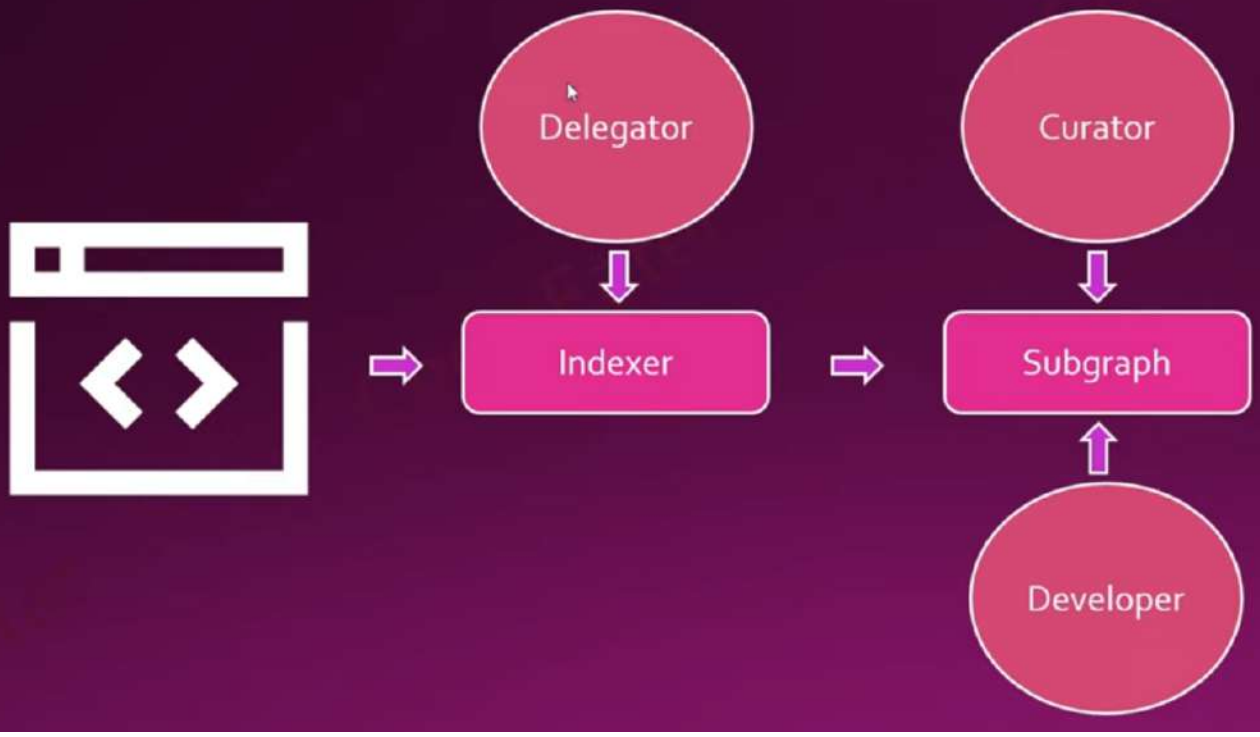
❓ Kya wo blockchain scan karta hai?
❌ NEVER

✅ They query **Subgraph**

🧠 Final brain-lock summary 🔒

> **Subgraph is NOT a contract.**
> **Subgraph is NOT blockchain.**
>
> **Subgraph = Indexed VIEW of blockchain data, built using events, stored in tables, queried with GraphQL.**

The Graph Working



🧠 One-line me poora diagram

> **Developer bolta hai kya chahiye → Curator batata hai kaunsa sahi hai → Indexer mehnat karta hai → Delegator support deta hai → User ko fast data milta hai**

Aur ye sab hota hai **The Graph** ke ecosystem me.

🎬 Is diagram ko ek MOVIE bana dete hain

Soch **ek restaurant business** 🍽️

👨‍💻 Developer (idea wala banda)

Role in diagram: bottom right

FEEL 👇

Developer bolta hai:

> "Mujhe Uniswap ke swaps ka fast data chahiye
> users, volume, timestamp sab dikhao"

👉 Developer:

- Subgraph likhta hai
- Events choose karta hai
- Schema define karta hai

🧠 **Developer = chef jo menu decide karta hai**

📜 Subgraph (menu + recipe)

Role in diagram: center right

Subgraph bolta hai:

> "Ye ingredients (events) lo
> aise chop karo
> aise plate me rakho"

📌 Subgraph = **rules + structure + data model**

🔥 **Ye wo cheez hai jo actually index hoti hai**

🧑‍⚖️ Curator (quality judge)

Role in diagram: top right

FEEL 👇

Curator aake bolta hai:

> "Haan bhai, ye subgraph useful hai
> log isko query karenge"

Wo kya karta hai?

- GRT stake karta hai
- Signal deta hai: "**Index THIS**"

🧠 **Curator = food critic / Zomato reviewer**

👉 Galat subgraph promote kiya?
💥 Paisa loss

🏗️ Indexer (real mazdoor 💪)

Role in diagram: center

FEEL 👇

Indexer bolta hai:

> "Theek hai, main ye subgraph run karta hoon"

Indexer:

- Graph Node chalata hai
- Blockchain read karta hai
- Events sunta hai 👂
- Data index karta hai
- Queries ka answer deta hai

🧠 **Indexer = restaurant kitchen + workers**

💥 Ye sabse heavy kaam karta hai

🤝 Delegator (silent investor)

Role in diagram: top middle

FEEL 👇

Delegator bolta hai:

> "Mere paas infra nahi
> par paisa hai"

Wo:

- Indexer ko GRT delegate karta hai
- Indexer ki power badhata hai
- Rewards me share leta hai

🧠 **Delegator = investor jo hotel me paisa lagata hai**

🖥️ Left side ka icon (Frontend / dApp)

Ye hai **actual USER WORLD**

- Website
- App
- Dashboard
- Analytics

User jab query karta hai:

```graphql
{
  swaps(first: 10) {
    amount
    user
  }
}
```

👉 Query seedha **Indexer** ke paas jaati hai
👉 Indexed data milta hai
👉 Fast response ⚡

🔁 Poora FLOW ek breath me

```mathematica
Developer
   ↓ (creates)
Subgraph
   ↑ (signals)
Curator
   ↓
Indexer ← Delegator (support)
   ↓
Frontend / dApp
```

💥 Sabse IMPORTANT FEEL (yaad rakh)

❌ Ye centralized system nahi hai

- Koi single server nahi
- Koi company ka control nahi

✅ Ye pure Web3 economics pe chalta hai

- Kaam = reward
- Galat kaam = loss
- Trust = stake

The Graph Roles

1. Indexer

An indexer is a participant in the Graph network who retrieves and indexes data from various blockchains, making it easily searchable and accessible through the Graph's query language, known as GraphQL. Indexers operate indexing nodes that store and organize data in the form of subgraphs. The more curated signal is there on the subgraph the indexer should choose that subgraph.
They play a crucial role in maintaining the decentralized indexing infrastructure of the Graph.

2. Delegator

A delegator is an individual or entity that holds Graph (GRT) tokens and delegates them to specific indexers. Delegators provide their stake to indexers to participate in the network and earn rewards based on the indexer's performance. Delegators play a role in securing the network and can choose to delegate their tokens to multiple indexers to diversify their participation. And make profit from the indexers.